Introduction To Programming

Python fundamentals

Universidad...

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Today's Goals

- Fundamentals
- Type of Variables
- Some Exercises

Fundamentals

Computers are obedient, not smart. They do exactly what you tell them to do, word for word, step by stepeven if your instructions lead to a error.
An **algorithm** is a precise, step-by-step set of instructions designed to solve a specific problem or complete a task.

Fundamentals

Computers Do What You Tell Them.

Think a computer of a kitchen assistant.If you tell it to "cook a complex dish". It will freeze because it doesn't know what that means. But if you give it a step-by-step recipe like: "chop 5 tomatoes, blend for 10 seconds, add 2 grams of salt", it will do it very quickly; because it moves at millions of operations per second.

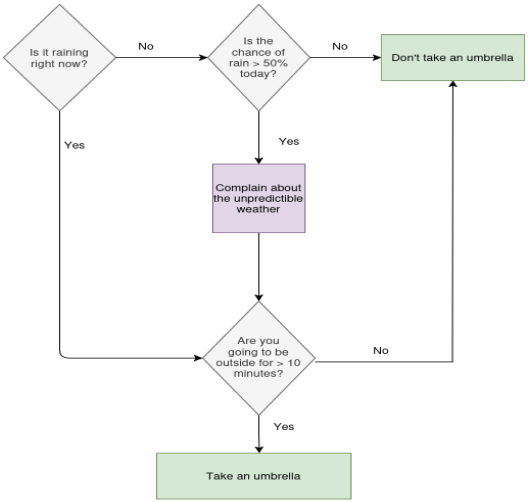
You Should Be Familiar With...

The Concept Of:

- Variable
- True/false statements
- Connecting statements (or,and, not)
- Making decisions (if ... then)
- Following flowcharts

If you shouldn't be familiar with the Flowcharts

You should be familiar with following a set of instructions and making branching desicions.



The structured program theorem

States that any computable algorithm can be expressed only with three logic control structures.

- Sequence or simple action: Executing one instruction after another in a specific, linear order.
- Conditional: valuating a boolean condition to "choose the correct path".
- Loop or iteration: Repeating a block of instructions as long as a certain condition remains true.

Variables

The = operator executes a **right-to-left assignment** (evaluating the right side first, then binding to the left name), not an algebraic equivalence.

Pitfalls & Invalid Code

Math equivalence: Treating = as static math logic (`a + x = b` throws `SyntaxError`).

Syntax violations: Starting with numbers (`1st = 5`), spaces, or overriding keywords (`print = 10`).

Single-letter cryptics: Using `x`, `t`, `p` without domain context.

Correct Model & Standards

Right-to-left binding: `x = x + 1` evaluates the new value first, then moves the name tag.

Valid naming: Letters or underscores (_), case-sensitive (`age` $\neq$ `Age`).

Style: Descriptive `snake_case` names (`circle_area`), $<$ 80 chars/line.

Exercise

"Allowed, Incorrect, Good Variables Names"

1.- customer_list
2.- print (where this word is a color other than black in Spyder)
3.- list_of_things_I_need_to_pick_up_from_the_store
4.- %score
5.- 1
6.- A

You'll be able to:

* Write code that creates various types of objects
* Write simple lines of codes to manipulate Python variables

Imagine...

A family made up of:

+ 4 people
+ 3 dogs
+ 2 cats

Every Person, cat, dog is a separate object (variable)
Characteristics are called data *attributes* or *values* for every type

Types

In the most programming languages few types of objects are basic building blocks for each language. These types are the 'primitives'. Python has five basic types (primatives) of objects:

i. Integers.
ii. Floats.
iii. Booleans.
iv. Strings.
v. A special type that represents the absence of a value .

Integers

Objects whose values are real whole numbers (1,0,3,5,-4,-100000).

You do on numbers what you do in math class.

Floats

An object of type floating point (float type) object whose values are decimal numbers. Example:

i. 0.0
ii. 2.3
iii. -22.8

Boolean

You need to work with more than just numbers. A boolean is a type that only has two possibles vales, *true* or *false*

Strings

A string is a sequence of characters surrounded by quotation marks. For example: 'hello', "We're 1!", 'Hi!'

Absence of a value

The value is None. Because in python everything is an object, this None also has a type, NoneType. It appear when in a list of variables the value is null.

converting between different types

When you are writing a program, each line of code is called a statement. if you want to know what kind of data type is an object, you should use ***type()***.
And if you want to change the type to another kind of data, use: int(), float(), str().

Exercise

Homework

write 2 programs in python. Start with a variable that contains the number of minutes (121,361). Your Program should show the results in the following ways:
the first program (121 minutes):
Hours
(number of hours)
Minutes
(number of minutes)
The second program (361 minutes): 361 minutes its equal to (hours) and (minutes) in one line.